

Extrapolation to the destabilizing tail

Experiment 02 · ddG_with_Boltz · raw-Δz features · gradient-boosted trees.

Trained only on mild mutations ($|\Delta\Delta G| < 1$ kcal/mol) and tested on the strongly destabilizing tail ($\Delta\Delta G > 2$), the predictor **does not extrapolate**: on the tail the correlation is ~ 0 and the predicted-vs-measured slope is ~ 0.02 — predictions saturate near the top of the training range while true values reach 5.7 kcal/mol. The model interpolates within the $\Delta\Delta G$ band it was trained on and cannot reach beyond it.

1. Motivation

The generalization study noted a predicted-vs-measured slope below 1 — the model under-predicts large effects (regression to the mean). This test pushes that to the extreme: by withholding the entire destabilizing tail from training, it distinguishes genuine extrapolation from interpolation within the training range.

2. Methods

A gradient-boosted-tree regressor on 256 raw-Δz features is fit on the mild set ($|\Delta\Delta G| < 1$) and evaluated on (a) held-out mild mutations — the in-distribution baseline — and (b) the destabilizing tail ($\Delta\Delta G > 2$). We report Pearson r, RMSE, MAE, the linear fit slope (measured = a + b·predicted), and how much of the true range the predictions cover.

3. Results

Split	n	Pearson r	RMSE	MAE	fit slope	pred max / true max
In-distribution (held-out $ \Delta\Delta G < 1$)	5022	0.565	0.35	0.28	0.29	0.81 / 1.00
Extrapolation tail ($\Delta\Delta G > 2$)	4435	0.088	2.28	2.18	0.02	0.92 / 5.68

Table 1. In-distribution baseline vs the extrapolation tail.

Extrapolation to the destabilizing tail — trained on mild mutations only

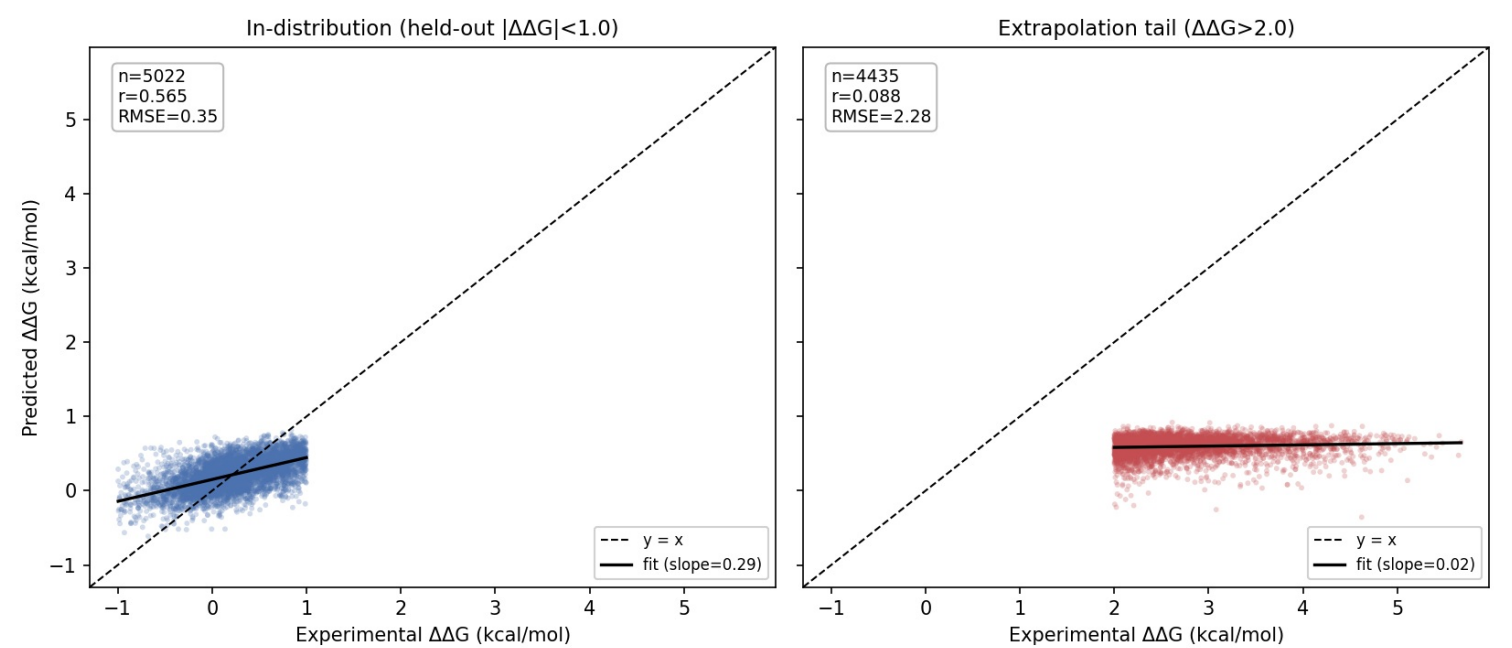


Figure 1. Predicted vs measured $\Delta\Delta G$ for the in-distribution baseline and the tail, with $y=x$ and fit lines. The tail is a flat cloud.

4. Discussion

On the tail the correlation collapses to 0.09 and the fit slope to 0.02 — essentially flat. Predictions cap at ~ 0.9 kcal/mol, the top of the training range, while true $\Delta\Delta G$ reaches 5.7. This is the regression-to-the-mean weakness in its extreme form: the model interpolates within the $\Delta\Delta G$ range it has seen and cannot extrapolate beyond it. The practical implication is direct — to rank or screen strongly destabilizing mutations, the training set must itself span that range; no amount of modeling recovers effect sizes absent from training.